

## For Loop and While Loop

```
for (initialization; condition; update) {  
    // body  
}
```

```
while (condition) {  
    // body  
}
```

### Sum of first N natural numbers using for loop

```
#include<stdio.h>  
void main()  
{  
    int n;  
    int i,sum=0;  
    printf("enter a number:");  
    scanf("%d",&n);  
    for(i=1;i<=n;i=i+1) //4<=3;i=4  
    {  
        sum=sum+i; //sum=3+3=6  
    }  
    printf("%d",sum);  
}
```

### Sum of first N natural numbers using while loop

```
#include<stdio.h>  
void main()  
{  
    int n;  
    int i=1,sum=0;  
    printf("enter a number:");  
    scanf("%d",&n);  
    while(i<=n)  
    {  
        sum=sum+i;  
        i=i+1;  
    }  
    printf("%d",sum);  
}
```

### Factorial of a number

```
#include<stdio.h>  
void main()  
{  
    int fact=1,n;  
    printf("enter a number:");  
    scanf("%d",&n);  
    if(n==0 || n==1)  
    {  
        printf("Factorial of %d is 1",n);  
    }  
    else  
    {
```

```

for(int i=2;i<=n;i=i+1)
{
    fact=fact*i;
}
printf("Factorial of %d is %d",n,fact);
}
}

```

### Reverse a number

```

#include<stdio.h>
void main()
{
    int n=125,k;
    while(n!=0)
    {
        k=n%10;
        n=n/10;
        printf("%d",k);

    }
}

```

### Reverse a number

```

#include<stdio.h>
void main()
{
    int n=125,k,sum=0;
    while(n!=0)
    {
        k=n%10;    //k=1
        sum=sum*10+k; //sum=521
        n=n/10;    //n=0
    }
    printf("%d",sum);
}

```

### Palindrome or not

```

#include<stdio.h>
void main()
{
    int n=1221,sum=0;
    int c=n;
    while(n!=0)
    {
        sum=sum*10+n%10; //sum=121
        n=n/10;    //n=0
    }
    if(c==sum)
        printf("Palindrome");
    else
        printf("Not a palindrome");
}

```

### Sum of digits

```
#include<stdio.h>
void main()
{
    int n=985,sum=0;
    int c=n;
    while(n!=0)
    {
        sum=sum+n%10; //sum=22
        n=n/10;      //n=0
    }
    printf("%d",sum);
}
```

## Challenge 4: Accepted



### 1. Even/Odd Numbers up to N

Write a C program to print all even and odd numbers from 1 up to a given number N.

Input: 10 → Output: Odd: 1 3 5 7 9 Even: 2 4 6 8 10

### 2. Count Digits in a Number

Write a C program to count the total number of digits in a given number.

Input: 45210 → Output: Number of digits = 5

### 3. Reverse a Number

Write a C program to reverse the digits of a given number.

Input: 1234 → Output: Reversed number = 4321

### 4. Sum of Digits

Write a C program to find the sum of all digits of a given number.

Input: 123 → Output: Sum of digits = 6

### 5. Armstrong Number Check

Write a C program to check whether a given 3-digit number is an Armstrong number.

Input: 153 → Output: 153 is an Armstrong number

### 6. Power of a Number ( $a^b$ ) using for/while loop

Write a C program to calculate the value of a number raised to a given power.

Input: Base = 2, Exponent = 5 → Output:  $2^5 = 32$

### **7. Palindrome Number Check**

Write a C program to check whether a given number is a palindrome.

Input: 121 → Output: 121 is a palindrome

### **8. Display Cube of Numbers Up to an Integer**

Write a program in C to display the cube of the number up to an integer.

Test Data :

Input number of terms : 5

Expected Output :

Number is : 1 and cube of the 1 is :1

Number is : 2 and cube of the 2 is :8

Number is : 3 and cube of the 3 is :27

Number is : 4 and cube of the 4 is :64

Number is : 5 and cube of the 5 is :125